

Martin Matak

Postdoctoral Researcher at the University of Utah
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<https://martinmatak.github.io>

EDUCATION

PhD in Computer Science @ University of Utah Aug 2019 – December 2024

- Advisor: Tucker Hermans
- Focus: Dexterous manipulation, robot perception, sim-to-real transfer, and real-world grasping of novel objects

MSc in Computer Science @ Vienna University of Technology (TU Wien) Oct 2016 – June 2019

- Focus: Attacks against Neural Networks

BSc in Computer Science @ University of Zagreb Sep 2013 – Jul 2016

- Thesis: *Data Processing with Technology Apache Spark*

RESEARCH EXPERIENCE

University of Utah

Postdoctoral Researcher Jan 2025 –

- Researching sim-to-real dexterous grasping for novel objects and real-world robot manipulation
- Explored technology translation through 100+ interviews with potential customers (NSF I-Corps)

NVIDIA, Seattle, US

Research Scientist Intern, Seattle Robotics Lab May 2022 – May 2024

- Developed DextrAH-G, a *pixels-to-action* dexterous arm-hand grasping policy for novel-object manipulation using RL and teacher-student distillation
- Demonstrated sim-to-real transfer on real robotic hardware; work formed the basis for a policy deployed on Boston Dynamics's Atlas [[blog post](#), [video](#)]

SELECTED PUBLICATIONS

- "First Plan Then Evaluate: Multi-Target Planning with Post-Planning Success Evaluation Improves Learning-Based Grasping Pipeline" [[website](#)]
[Martin Matak](#), Mohanraj Devendran Shanthi, Karl Van Wyk, Tucker Hermans
IEEE Robotics and Automation Letters (RA-L) 2026
- "DextrAH-G: Pixels-to-Action Dexterous Arm-Hand Grasping with Geometric Fabrics" [[website](#)]
[Martin Matak*](#), Tyler Ga Wei Lum*, Viktor Makovychuk, Ankur Handa, Tucker Hermans, Nathan D. Ratliff, Karl Van Wyk
Conference on Robot Learning (CoRL) 2024
- "Planning Visual-Tactile Precision Grasps via Complementary Use of Vision and Touch" [[PDF](#), [Video](#)]
[Martin Matak](#) and Tucker Hermans
IEEE Robotics and Automation Letters (RA-L) 2023

OTHER EXPERIENCE

Deloitte Digital, Vienna, Austria

Software Engineer May 2018 – July 2019

- Backend (Java, Oracle) for a loyalty program for grocery stores
- Collaborated directly with customers to inform product development

Austrian Institute of Technology (AIT), Vienna, Austria

Data Science Intern

Oct 2017 – Feb 2018

- Investigated linkability of *monero* cryptocurrency (Scala, Google Cloud).

CROZ d.o.o., Zagreb, Croatia

Software Engineering Intern

Jul 2016 – Oct 2016

- Worked on graph search through NLP (before LLMs)

OTHER PUBLICATIONS

- “Robust Bayesian Scene Reconstruction by Leveraging Retrieval-Augmented Priors” [\[website\]](#)
Herbert Wright, Weiming Zhi, [Martin Matak](#), Matthew Johnson-Roberson, Tucker Hermans
IEEE Robotics and Automation Letters (RA-L) 2026, presented at ICRA 2026.
- “23 DoF Grasping Policies from a Raw Point Cloud” [\[PDF\]](#)
[Martin Matak](#), Karl Van Wyk, Tucker Hermans
IEEE International Conference on Robotics and Automation (ICRA) Workshop on Geometric Representations 2023
- “Comparing Piezoresistive Substrates for Tactile Sensing in Dexterous Hands” [\[PDF\]](#)
Rebecca Miles, [Martin Matak](#), Mohanraj Devendran Shanthi, Darrin Young, Tucker Hermans
Preprint
- “Learning Continuous 3D Reconstructions for Geometrically Aware Grasping” [\[PDF\]](#)
Mark Van der Merwe, Qingkai Lu, Balakumar Sundaralingam, [Martin Matak](#), Tucker Hermans
IEEE International Conference on Robotics and Automation (ICRA) 2020